# Supplementary material

From routine cine CMR to regional myocardial dysfunction

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11 October 2026

## Supplementary Methods: literature search, screening and evidence handling

**Review type and records.** This is a critical narrative review, not a systematic review. No registered protocol, duplicate screening, meta-analysis or PRISMA flow is claimed. The supplied history describes a structured search, citation expansion and subsequent targeted updates. The preserved query log and extraction notes establish part of that history; missing original exports and record-level decisions prevent full reconstruction of the reported screening process.

**Core set and citation expansion.** The supplied history identifies a core set including DeepStrain and its evaluations [1–3], StrainNet [4], multivendor and material-sensitive comparisons [5–8], consensus documents [9–12] and reviews [13–17]. It describes following references and citing articles, manually adding guideline and resource records, and stopping when no method family, comparison axis or main conclusion changed. The round-by-round expansion record is unavailable, so the stopping rule is retained as a historical description rather than a reproduced saturation assessment.

**Historical structured search.** The surviving query log records PubMed, Europe PMC and OpenAlex queries on 3 September 2026 with a publication cutoff of 1 September 2026. Six concept families were queried in each database: regional cine strain after infarction; known-motion or digital-phantom validation; prior or regularisation perturbation; conditional recoverability; failure-aware or selective prediction; and regional localisation, spillover or attenuation. The 18 exact queries, hit counts and dates are preserved in `supplementary/historical/gap-search-query-log.csv`. Recorded hit totals are 41, 244 and 508, respectively. Retrieval was capped at 100 records per query, truncating the broad post-infarction query in Europe PMC (239 hits) and OpenAlex (473 hits). The per-query returned counts sum to 281. The historical notes report 253 unique records after deduplication, but the original exports and screening file needed to recompute that count are not present.

**Historical screening and retention.** The notes describe one software-assisted title-and-abstract screening pass and no full-text exclusion stage. The intended design combined prescribed regional abnormality, known motion and controlled estimator changes; records addressing part of that design were also retained. The reported counts are 21 retained and 232 set aside, followed by two additions on 9 September under the same publication cutoff (Righetti and colleagues, 2025; Yang and colleagues, 2025, preprint). A preserved 23-row near-neighbour matrix includes two versions of MulViMotion and was described as 22 studies. These are inherited counts and classifications, not a new screening result. Individual decisions, deduplication and the historical conclusion that no retained study combined every design element cannot be re-audited from the

available files. That joint criterion would not in any event establish absence of studies addressing a simpler subset. The revised main text therefore treats graded focal recovery as a proposed priority, without using this record to claim field-wide absence.

**October updates and present verification.** The supplied narrative dates a targeted verification pass to 8 October 2026; its bibliographic ledger is named `evidence/crossref_verification_20261009.csv`. Sources published after 1 September, including the September Jahromi study and October IG-BiReg record, belong to this later update, whose complete query and inclusion history is unavailable. On 11 October, central numerical and methodological claims were checked against accessible primary-source sections and official resource pages. The separate dated addendum records the actual targeted queries, direct URL retrievals, access failures and reading depth. It is not a rerun of the historical search. The existing independent identity audit covers 94 references; identity retrieval does not certify every detailed claim or a full-text reading of every source.

**Reading depth and unresolved detail.** The inherited 135-row claim ledger labels 75 entries full text, 56 abstract and 4 official record. It covers 69 citation keys, which do not match the complete current 94-reference set; many locators refer to source files absent from this package. Those labels are retained as historical provenance, not current verification. The updated claim and coverage tables distinguish inspected full-text sections, abstracts, metadata, official records, inherited extraction and failed access. Supplementary Table S3 preserves method-specific summaries at those available depths. NV denotes detail not verified by this review, rather than a claim that the source did not report it. A bibliographic-only entry is not assigned a validation result.

**Software assistance.** The supplied history attributes search and drafting assistance to OpenAI Codex and Claude Code (Anthropic). The present integration used OpenAI Codex to compare the three review jobs, check selected primary-source passages, reconcile the latest figure package and compile the manuscript. An earlier fresh-context AI review evaluated an intermediate draft; subsequent targeted rechecks and this integration are not independent human screening or clinical review. Figure sources were edited programmatically and rendered with headless Chromium; Tectonic compiled the multi-file document. Author verification and responsibility remain pending, as stated in the main declarations.

## Supplementary Note 1: strain definitions and comparison traps

Let $\boldsymbol{\varphi} : \Omega_0 \to \Omega_t$ map a reference position $\boldsymbol{X}$ to its deformed position $\boldsymbol{x}$. The displacement is

$$\boldsymbol{u}(\boldsymbol{X}, t) = \boldsymbol{\varphi}(\boldsymbol{X}, t) - \boldsymbol{X}, \tag{1}$$

and the deformation gradient is

$$\boldsymbol{F} = \frac{\partial \boldsymbol{\varphi}}{\partial \boldsymbol{X}} = \boldsymbol{I} + \frac{\partial \boldsymbol{u}}{\partial \boldsymbol{X}}. \tag{2}$$

Common finite-strain measures include the right Cauchy–Green tensor $\boldsymbol{C} = \boldsymbol{F}^{\mathsf{T}}\boldsymbol{F}$, Green–Lagrange strain $\boldsymbol{E} = (\boldsymbol{C} - \boldsymbol{I})/2$, and logarithmic strain obtained from a stretch tensor. In one dimension, engineering strain $e = (\ell - \ell_0)/\ell_0$ and Green–Lagrange strain satisfy $E = e + e^2/2$. Numerically comparing values without converting the definitions can create an apparent method bias [7].

Cardiac components require a coordinate frame. Circumferential, radial and longitudinal axes can be defined at the reference state, updated through time or constructed from a ventricular centreline. Local axes become unstable near the apex and can vary with segmentation. A transmural average differs from endocardial

strain because deformation varies through the wall. Two-dimensional short-axis analyses omit longitudinal and through-plane derivatives; a reported radial or circumferential component is consequently conditional on the 2D model.

Peak definitions also differ. If $E_s(t)$ is the segment-average strain curve, $\min_t E_s(t)$ is generally not equal to the spatial average of pointwise minima. "End-systolic", "peak systolic" and "peak over the whole cardiac cycle" may diverge in dyssynchrony or post-systolic shortening. A reproducible comparison must fix the order of differentiation, projection, spatial averaging and temporal peak selection. Supplementary Figure S1 illustrates these four traps.

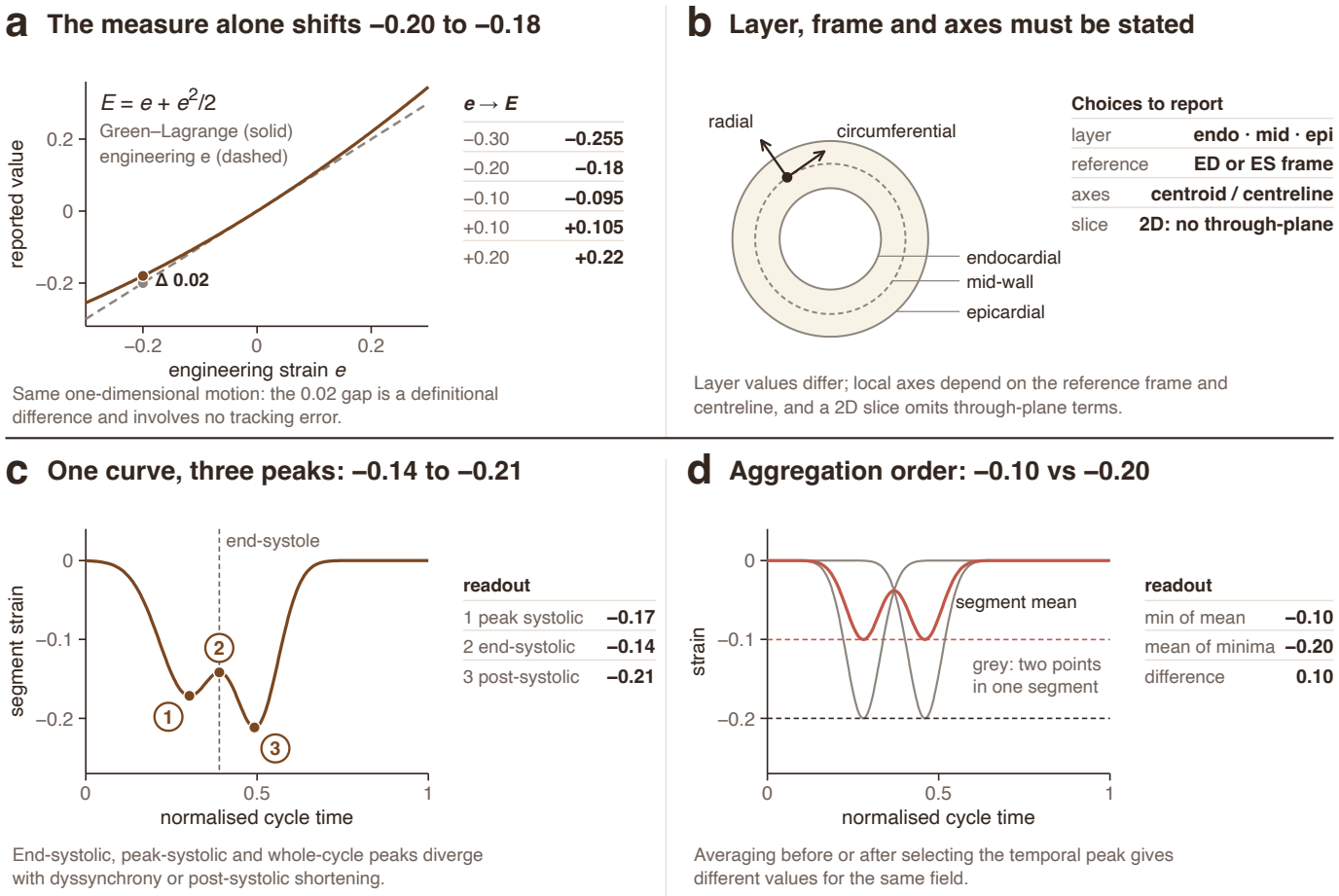



**Figure S1.** The same motion gives different numbers under different definitions. (a) Engineering strain $e$ and Green–Lagrange strain $E = e + e^2/2$ differ by 0.02 at $e = -0.20$; the table converts five values. (b) Endocardial, mid-wall and epicardial layers and local radial and circumferential axes, with the choices a report must state. (c) One segmental curve read as peak systolic (−0.17), end-systolic (−0.14) and post-systolic peak (−0.21). (d) The minimum of a segment-mean curve (−0.10) differs from the mean of pointwise minima (−0.20). Conceptual and analytic schematics with no measured scale; printed values are computed from the drawn curves.

## Supplementary Note 2: validation targets

The claim boundaries for common validation targets are summarised in Figure 5a of the main text.

A validation hierarchy should not be interpreted as a single ranking. Simulation supplies exact truth but imperfect realism. DENSE and tagging provide material-sensitive in-vivo information but have sequence- and processing-specific error. Scan–rescan establishes precision. LGE and outcomes test biological or clinical relationships. A mature claim often needs several targets because their weaknesses are complementary.

When two acquisitions are compared, acquisition registration is part of the reference. Slice position, breath-

hold, loading and heart-rate differences can create apparent algorithm error. Where reference processing includes smoothing or temporal fitting, its effective resolution should be reported. Segment-level analyses require participant-aware uncertainty estimates because segments from one heart are correlated.

## Supplementary Note 3: error attributes and mapping validity

The distinct spatial, temporal and field-validity failure modes are illustrated in Supplementary Figure S2.

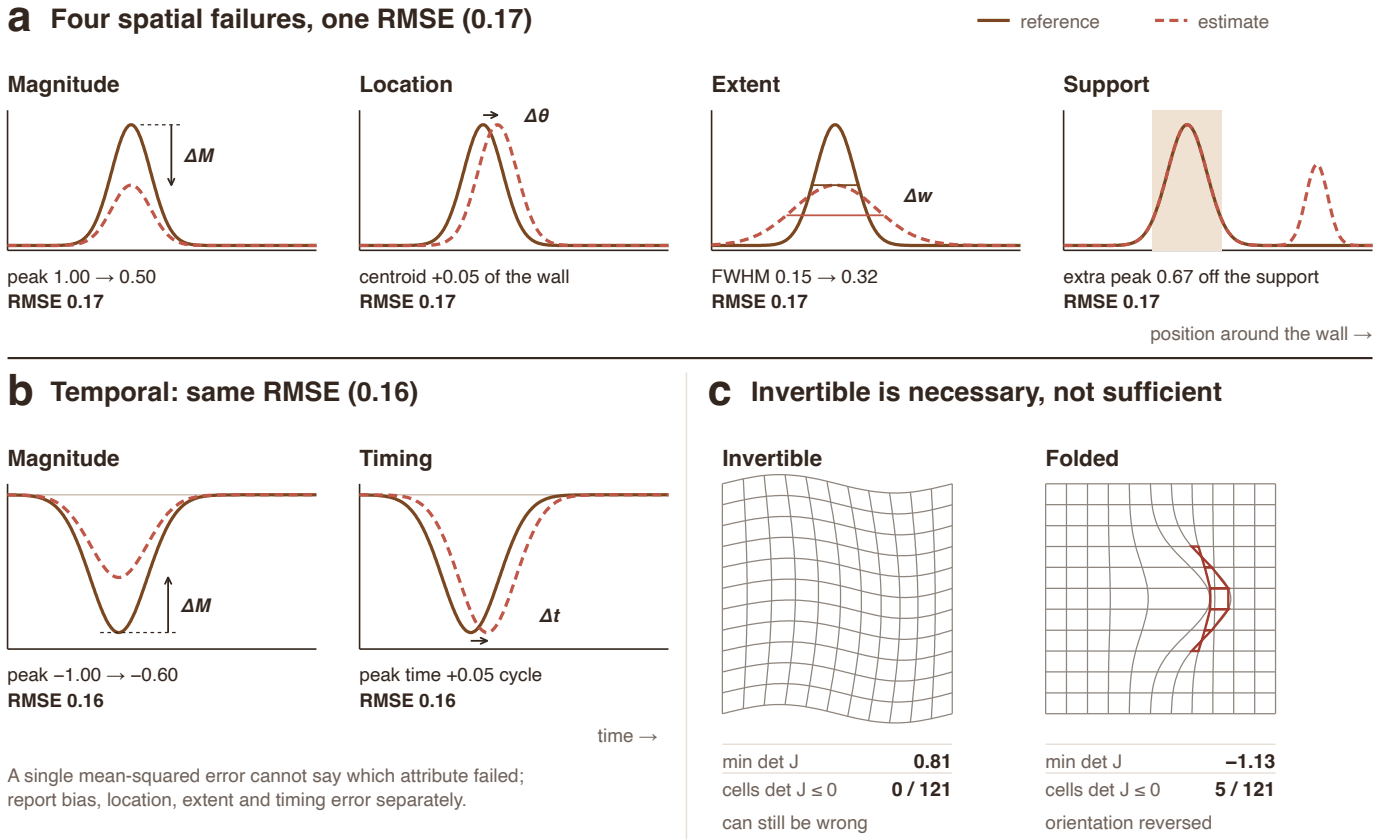



**Figure S2.** Regional error attributes and mapping validity. (a) Reference and estimate profiles separating magnitude, location, extent and support errors. (b) Temporal magnitude and timing errors. In a, and again in b, every estimate has the same root-mean-square error against its reference (0.17 and 0.16), so a single mean-squared error cannot tell these failures apart. (c) A smooth invertible mapping (minimum $\det J = 0.81$), which can still be wrong, and a folded mapping with $\det J \leq 0$ in 5 of 121 outlined cells (minimum $-1.13$). Prescribed schematics with no measured scale.

Let a prescribed abnormality field be $g_{\mathrm{ref}}(\boldsymbol{x}, t)$ and an estimate be $g_{\mathrm{est}}(\boldsymbol{x}, t)$. A single mean-squared error can conceal clinically different failures. Attribute-specific losses may include:

- magnitude error in an agreed region;
- distance between abnormality centroids or peaks;
- width, area or overlap error of a thresholded abnormal region;
- onset, peak-time or curve-alignment error; and
- error conditional on the reference support and a sensitivity analysis over the threshold.

Field validity is related but distinct. A positive Jacobian determinant is a useful check against local folding, while near-incompressibility may be physiologically motivated. Neither proves accurate correspondence. A

smooth, invertible mapping can be wrong, and a reference generated by an approximate 2D model can violate three-dimensional tissue behaviour. Validity metrics should accompany, not replace, regional accuracy metrics.

## Supplementary Note 4: learning target and control point

DeepStrain and StrainNet illustrate why the target matters. DeepStrain demonstrated a complete automated workflow with segmentation, motion estimation, volumetric agreement, reference ranges and repeatability analyses [1]. Its cine motion network was not independently validated against dense material displacement for every regional claim. StrainNet used DENSE phase-derived displacement as a supervised target, but it reported two distinct evaluations [4]. In the 62-person internal participant-held-out test set, its 0.75 ± 0.35-mm pixelwise EPE was obtained when DENSE-magnitude contours, not cine contours, were supplied to the network and compared with DENSE displacement over myocardium and time. Separately, bSSFP cine feature-tracking contours were used for the end-systolic global and AHA-segmental circumferential-strain comparison in 59 test participants. Thus the EPE and cine-contour ICCs must not be fused into one "cine displacement" result; population, contour source, registration, direction, aggregation and time point still bound the claim.

Training-time and inference-time controls must also be distinguished. Changing a loss weight usually requires matched retraining, and varying it only at test time has no effect unless the trained model exposes that parameter. Post-processing strength or a biomechanical solve, by contrast, can be a genuine inference-time control. Evaluation should compare like with like, hold splits and estimands fixed, and keep final test cases out of model or rule selection.

Supplementary Figure S3 shows the two control points and their shared held-out evaluation contract.

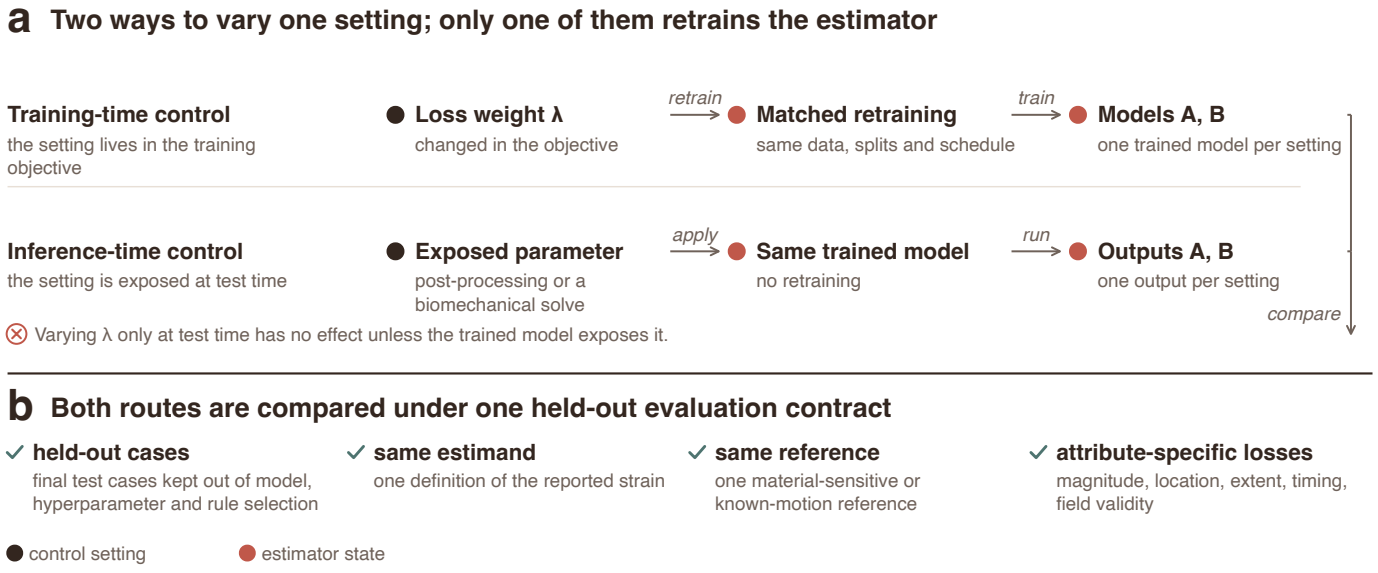



**Figure S3.** Training-time versus inference-time controls. A training-loss weight requires matched retraining; a test-time parameter can be varied only when the implementation exposes it. Candidate models or outputs are compared on the same held-out cases under the same estimand, reference and attribute-specific losses. Conceptual schematic.

## Supplementary Note 5: pre-specified evaluation template

For each intended-use claim, the following items can be completed before the final test set is analysed:

| Item | Pre-specified content |
|---|---|
| Intended use | Clinical/research decision; user; consequence of false regional call |
| Population | Disease, age, rhythm, prior treatment, exclusions and recruitment setting |
| Acquisition | Scanner/vendor, field strength, sequence, views, voxel size, slice gap, temporal resolution, acceleration/reconstruction |
| Abnormality | Direction, magnitude, location, width/extent, transmurality and timing range |
| Estimator | Name/version, inputs, user interaction, representation and regularisation |
| Estimand | Reference, strain measure, axes, layer, dimensionality, region, temporal summary and aggregation |
| Reference | Simulation/phantom, DENSE, tagging, landmarks, tissue target or outcome; registration and uncertainty |
| Primary loss | Attribute-specific error at subject level |
| Tolerance | Clinically/technically justified threshold and sensitivity analysis |
| Splits | Participant-, centre- and time-aware separation; no final-test selection |
| Subgroups | Scanner, centre, image quality, pathology, wall thickness, abnormality scale and demographic groups |
| Failure policy | Exclusion, imputation, quality control or abstention; observable signals and coverage |
| Statistics | Bias/limits, confidence intervals, repeated-measures model, multiplicity and missingness |

## Supplementary Note 6: software, versioning and regulatory context

Feature-tracking and strain tools change over time. Regulatory clearance may cover a broad software application or specified outputs but does not establish material-motion accuracy for every regional analysis. Product webpages document current availability, not historical algorithms. Studies should therefore record software name, exact version, analysis module, contour convention, manual corrections and any export transformation. Reference intervals and serial thresholds should not be transferred across products without a bridging study [5, 6, 18].

The source package does not bundle historical FDA summaries or manufacturer webpages and therefore does not claim that such records were independently audited in this correction pass. Bibliographic citations and dated review notes support only the limited interpretation above; they are not a continuously current market survey. Authors should verify any product indication, clearance, version and jurisdiction against the live primary record before making a clinical recommendation.

## Supplementary Note 7: reconstruction and dataset scope

Accelerated reconstruction can alter temporal sharpness, noise texture and edge appearance, all of which may affect downstream motion estimation even when conventional image-quality scores remain acceptable. Primary evidence is encouraging but endpoint-specific. In REGAIN, global feature-tracking strain from 49 participants showed near-zero mean differences between separately acquired accelerated and GRAPPA

breath-hold cine, with non-zero limits of agreement [19]. CineVN's paired 18-volunteer comparison found the smallest strain bias among the tested accelerated reconstructions, but all accelerated protocols retained systematic peak radial/circumferential bias; the 46-patient demonstration had no conventional reference and included segmentation-related slice exclusions [20]. In ten healthy volunteers, model-based DLR reduced the dispersion of global circumferential strain differences from full sampling at high reduction factors relative to compressed sensing, while bias and radial underestimation remained [21]. DENT improved image-phase interpolation and reader-rated temporal presentation across a large multicentre development set and prospective testing, but it did not evaluate myocardial strain or validate its deformation vectors as tissue motion [22]. These measured findings motivate joint reconstruction–estimator tests; they neither show nor imply that deep learning erases focal abnormalities.

A reconstruction study that agrees on volumes, global strain or image appearance does not automatically validate regional correspondence, lesion location, spatial extent or timing. Ideally, the reconstruction and motion estimator are evaluated jointly against a material-sensitive or prescribed-motion reference, with the fully sampled or conventional reconstruction treated as a comparator that carries its own error. Temporal blurring, sharpening, alias suppression and hallucination are candidate mechanisms, not established explanations for a regional result unless directly tested.

CMRxRecon2025 is useful for reconstruction robustness because the official site lists multiple centres, vendors, disease groups and a broad set of CMR modalities [23]. The official description does not provide material-motion ground truth, and it does not justify assuming that cine and LGE are paired for every participant. The 2024 challenge included facts and tasks that should not be transferred to 2025. Dataset labels, access, pairing, licensing and intended task must each be verified before use.

## Supplementary Note 8: study-level interpretation examples

| Study | Strongest direct support | Important positive evidence | What remains open |
|---|---|---|---|
| StrainNet [4] | DENSE-contour pixel-wise displacement EPE; separately, cine-contour end-systolic global/segmental Ecc | Higher global and segmental Ecc agreement than conventional FT in 59 internal-test participants | EPE is not a cine-contour result; focal location/extent, other directions, external domains and reference uncertainty |
| Militaru et al. [5] | Product-specific global/regional agreement with tagging and LGE discrimination | Strong global LS/CS agreement for tested products | Regional/radial interchangeability and material truth |
| Kihlberg et al. [8] | Scar-discrimination performance in suspected CAD | DENSE/tagging outperformed FT for high-transmurality LGE segments | Pointwise motion error and other diseases/thresholds |
| Strain-8 [24] | Same-day precision by modality/direction/field strength | Several global measures had good or excellent repeatability | Accuracy, disease cohorts and longer-term reproducibility |
| MRXCAT2.0 [3] | Technical capacity/error under generated cases | Known motion paired with realistic-looking cine | Unmodelled anatomy, artefact, scanner and disease complexity |

| | | | |
|---|---|---|---|
| Jahromi et al. [25] | Phantom and healthy-volunteer component-specific comparison | Circumferential recovery was stronger than other directions in tested setting | Pathology, base/apex, 3D effects and clinical thresholds |

# Supplementary Table S1: empirical validation contracts

**Table S1.** Detailed measurement contracts for selected empirical studies. Unextracted or unverified items describe the limits of this review, not absence from the original study.

| Study and denominator | Population, input and analysis unit | Output, scale, layer, reference state and time | Reference, statistic and result | Directly supported inference |
|---|---|---|---|---|
| Wehner et al. [7]; 88 participants; 186 paired slices; torsion and dyssynchrony subset 38 | Healthy and heterogeneous cardiac diagnoses from two centres; bSSFP plus matched DENSE; subject and slice | Transmural whole-slice base/mid/apex peak Green–Lagrange Ecc; four-chamber longitudinal strain; torsion and curve-delay dyssynchrony, ED reference | DENSE; mid Ecc bias −0.4%, CoV 11%; base bias −4.0%, CoV 18%; apex bias −2.4%, CoV 15%; dyssynchrony bias 26 ms, CoV 76% | Mid-slice peak Ecc agreement under matched definitions; poorer basal/apical magnitude and temporal synchrony agreement |
| Cao et al. [6]; 95 recruited, 87 analysed (77 cardiomyopathy, 10 controls) | Matched tagging, DENSE and bSSFP analysed by three FT products; subject and one matched mid slice/4-chamber | Peak Lagrangian Ecc and Err in one mid short-axis slice; Ell in four-chamber; systolic peak | HARP tagging; DENSE Ecc ICC 0.778, bias 0.155 percentage points, 95% limits −6.166 to 6.476; all methods had poor Err agreement | Relatively close Ecc but technique-dependent Err/Ell; no focal location, extent or dense-field endpoint |
| Militaru et al. [5]; 61 participants | 18 healthy, 18 chronic MI, 15 DCM, 10 LV hypertrophy; bSSFP with three FT products, 3 T; subject and AHA segment | Global and 16-segment transmural peak-systolic Lagrangian LS, CS and RS; ED reference | HARP tagging; global ICC 0.92–0.94 LS, 0.88–0.91 CS; intervendor ICC 0.10–0.81 for global RS (FT–FT, not FT–tagging); LGE discrimination tested in MI | Strong global LS/CS agreement for tested products; regional/radial values and scar discrimination are product dependent |
| Kihlberg et al. [8]; 116 patients | Suspected CAD; DENSE, tagging, bSSFP FT and LGE; subject and segment | ROC endpoint: end-systolic segmental circumferential strain; other global/segmental strain and torsion analyses were also reported | LGE > 50% transmurality; segment AUC 0.87 DENSE, 0.83 tagging, 0.66 FT; sensitivity at 80% specificity 82%, 71%, 35% | Technique-specific discrimination of high-transmurality scar; not a dense displacement ranking |
| StrainNet [4]; 305 total, 243 development/670 sections, 62 internal test/190 sections; cine analysis 59 | Patients, healthy adults and healthy children/adolescents; DENSE magnitude contours for displacement test, separately bSSFP cine FT contours; pixel, subject and AHA segment | Intramyocardial displacement over frames; separately end-systolic global/segmental circumferential Green–Lagrange strain, ED reference | DENSE phase-derived displacement; EPE $0.75 \pm 0.35$ mm with DENSE contours; cine Ecc ICC 0.87 global/0.75 segmental versus FT 0.72/0.48; segmental LoA −9.16 to 8.07 percentage points | Positive internal-test displacement performance under DENSE-contour input and bounded cine end-systolic Ecc agreement; the two tests are not interchangeable |
| Strain-8 [24]; 20 healthy volunteers | Same-day rescan using cine, tagging and DENSE at 1.5/3 T plus 2D/3D echo; subject and segment | Global and AHA-segmental frame-wise peak strain; modality-specific layer/definition | Repeat scan; CoV, ICC and Bland–Altman; global CoV ≤ 20% across protocols, segmental values frequently > 20%, especially longitudinal/radial | Precision and minimum detectable change under controlled same-day conditions; no accuracy or disease-cohort claim |

| Study and denominator | Population, input and analysis unit | Output, scale, layer, reference state and time | Reference, statistic and result | Directly supported inference |
|---|---|---|---|---|
| Jahromi et al. [25]; deforming phantom and 40 healthy volunteers | One-slice cine model versus one-/two-slice DENSE models; subject, slice and endocardial/epicardial layer | Slice-wise Ecc, Ell and Err through the normalised cycle; midventricular; model-specific reference | Full-text results: peak median systolic endocardial Ecc, cine minus two-slice DENSE −2.7 percentage points; TOST within selected ±3.5-point margin; slice-clustering treatment unclear | Endocardial Ecc equivalence in healthy mid-slice setting; not base/apex, disease or focal lesion validation |

For StrainNet, the Results prose reports global cine–DENSE limits of agreement of −5.74 to 4.72 percentage points, whereas Table 2 gives bias −0.51 with a 95% limits half-width of 5.15 percentage points. These do not reproduce exactly the same interval. The global interval in the main text follows the source prose; the segmental interval is consistent across the two presentations. This small source discrepancy is retained rather than silently reconciled [4].

## Supplementary Table S2: validation-resource details

**Table S2.** Validation resources and reference conditions. Descriptions combine the supplied October records and targeted checks on 11 October 2026; unresolved access is retained. The phrase not file-verified means that no subject-level archive was downloaded or processed in this review.

| Resource | Nature, scale and population | Modalities, coverage and pairing | Motion reference and evaluable scale | Access, processing requirement, uncertainty and domain boundary |
|---|---|---|---|---|
| STRAUS [26, 27] | Synthetic: 18 cases generated from 3 volunteer templates, each with healthy, LBBB, LCX, LAD, RCA and RCA2 states | For each case: 3D cine MR, three tagged-MR sequences and 3DUS over the cardiac cycle; 90 sequences/2700 volumes reported | Framewise reference meshes provide dense correspondence; the paper reports regional engineering strain | Public/no login reported; MHD/RAW images and VTK meshes are ready files but require matched coordinate/time/reference handling; imposed mechanics, three anatomies and simulated appearance bound transfer |
| Stanford Cardiac Performance MRI [28] | Human: laboratory page 51 healthy; DOI metadata 55 total (51 healthy, 4 aortic stenosis), acquired 2023–2025 | 3 T whole-LV bSSFP cine; DENSE at corresponding locations with xyz encoding; three mid-LV tagged slices; T1 and cDTI described | DENSE phase and tagging can support in-vivo displacement/strain comparisons after processing; whole-LV cine coverage does not imply whole-LV tagged reference | Stanford Digital Repository, ODbL 1.0; subject ZIP DICOMs; not file-verified here; requires reconstruction/phase unwrapping, segmentation, registration and estimand alignment; cohort total is not a usable-pair count; Jahromi reports using this resource, but subject-file linkage is not verified |
| CMAC [29, 30] | Dynamic phantom and 15 healthy volunteers; challenge database | Cine SSFP, 3D tagged MRI and 3DUS; whole-cycle 3DTAG and cine with modality registration described | 12 landmarks (four walls at three LV levels) tracked by two observers in 3DTAG; sparse point error, not dense field truth | Open access with attribution; median inter-observer variability 0.77 mm phantom and 0.84 mm volunteers; healthy/sparse/legacy acquisition domain |
| MRXCAT [31] | Synthetic generator based on XCAT; not a fixed participant cohort | Cine and first-pass perfusion with cardiac/respiratory motion, multiple coils, noise, trajectories and undersampling | Prescribed XCAT motion and fully sampled simulation support sampling/reconstruction and known-motion tests | MATLAB framework available for non-commercial research by request; XCAT access and self-generation required; signal, texture, partial-volume and anatomy assumptions limit realism |

| Resource | Nature, scale and population | Modalities, coverage and pairing | Motion reference and evaluable scale | Access, processing requirement, uncertainty and domain boundary |
|---|---|---|---|---|
| MRXCAT 2.0 [3, 32] | Synthetic generator, not a fixed cohort; healthy, infarct, DCM and HCM exemplars | bSSFP-like cine generated from statistical LV shape, biophysical function and texturised tissue labels | Known simulated function/strain enables voxel/mesh-scale capacity and estimator tests | Repository cited by the paper; live code/example access was blocked during this check; generation and dependencies require verification; generated cases are not independent patients and model/texture coverage bounds transfer |
| CMRxRecon 2025 [23] | Human multicentre, multivendor, multidisease reconstruction challenge/resource; exact motion-reference denominator not applicable | Cine plus mapping, contrast-weighted, black-blood, phase-contrast, LGE and perfusion described; pairing varies by task/release | No declared dense material-motion truth; can assess reconstruction robustness and downstream domain shift if endpoints are added | Challenge terms and released reconstruction/QC files; disease labels are not motion truth and cine–LGE pairing cannot be assumed for every participant |

# Supplementary Table S3: cine-CMR motion estimators by year

**Table S3.** Motion and strain estimators discussed in the main text, in approximate chronological order within each family. Summaries retain the source-specific input, representation and available validation description. NV means not verified at the available reading depth; it is not an absence claim. Most lineage entries use abstracts; full-text corrections to central studies are identified in the main text and evidence ledger. Rows are not ranked and differ in status (journal, conference, workshop, preprint); the year/source column states which is which. A short validation entry usually means the method was evaluated for the claim its own authors made, not that it was tested and found wanting for this review's question.

| Method | Year; source | Input; supervision | Prior or representation | Reported data and validation | Evaluation on pathology or focal abnormality |
|---|---|---|---|---|---|
| *Conventional estimators and comparators* | | | | | |
| Boundary feature tracking | Family [33] | Cine plus endo/epicardial contours; per-case tracking | Contour continuity and propagation; interior motion largely interpolated | Many clinical cohorts; inter-software/version agreement, scan–rescan, tagging/DENSE comparison, clinical association | Many disease cohorts; regional and radial agreement consistently weaker than global [5, 24, 34] |
| Free-form deformation (FFD) | 1999; IEEE TMI [35] | Image pair; none | Affine plus B-spline lattice with smoothness cost; NMI data term | Contrast-enhanced breast MRI; comparison with rigid/affine registration | Not applicable (non-cardiac origin); transformation model reused by later cardiac work |
| Harmonic phase (HARP) | 2000; IEEE TMI [36] | SPAMM-tagged MR (comparator, not cine); none | Phase of an isolated spectral peak | Real and simulated tagged images; small-motion displacement and 2D strain | Not applicable; material-sensitive comparator |
| Early clinical feature tracking | 2009–2013 [37–39] | Cine plus contours; vendor feature tracking, per case | Proprietary feature tracking | 30 post-STEMI patients (scar association); 191 DMD patients and 42 controls versus HARP, mid-LV whole-slice Ecc; 145 healthy volunteers with tagging in 20 (abstract level) | Scar association and whole-slice/global agreement; focal localisation endpoint NV |
| Groupwise $n$D+$t$ B-splines | 2011; Med Image Anal [40] | Whole dynamic sequence; none | Spatio-temporal B-spline, optional cyclic constraint, no fixed reference | Synthetic and 3D+$t$ lung/cardiac CT; sub-voxel accuracy; cardiac CTA volume curves with mean deviation about 3% from manual curves (abstract) | NV |

| Method | Year; source | Input; supervision | Prior or representation | Reported data and validation | Evaluation on pathology or focal abnormality |
|---|---|---|---|---|---|
| Laplace-based contour mapping | 2015; Int J CARS [41] | Cine contours; none | Laplace equation between successive contours | Synthetic data and in-vivo feasibility (abstract level) | NV |
| Variational data assimilation | 2016; IEEE TMI [42] | Cine sequence; none | Contraction model coupled to image observations | Known-motion synthetic data; 47 patients at three slice levels; tagging strain ICC above 0.8 and clinical-score associations (abstract) | NV as a prescribed focal experiment |
| Gradient-based optical flow | 2019; Med Image Anal [43] | Cine and tagged MR; none | Intensity and shape-feature invariance, iteratively optimised | Realistic simulations and preliminary clinical data (abstract level) | NV |
| Automated segmentation plus spline tracking | 2017; FIMH [44] | Cine; supervised segmentation, unsupervised tracking | Learned segmentation feeding a conventional tracker | 42 HCM patients and 21 controls; pixel accuracy about 97% against manual segmentation | Group separation HCM versus controls; material-motion reference and focal experiment NV |
| Automated segmentation plus feature tracking | 2018; ISBI [45] | Short- and long-axis cine; supervised segmentation | CNN segmentation followed by conventional feature tracking | 40 healthy volunteers and 40 ischaemic patients; strain agreement with cvi42 (abstract) | Inter-method agreement; material-motion reference and focal experiment NV |
| Anatomy-Aware Tracker | 2020; MLMI [46] | Long-axis cine; weak shape supervision | Convolutional VAE of realistic shapes refining a dense tracker | Tracking versus other methods; qualitative shape realism | NV |
| Groupwise locally low-rank | 2024; BMC Med Imaging [47] | Whole cine sequence; none | Groupwise registration minimising patchwise signal rank | Simulation ($n = 20$), public cine ($n = 100$), in-house tagging patients ($n = 16$); point, voxelwise and global strain error; late-diastolic drift; tagging radial-strain correlation | Paired tagging in patients; disease composition NV; prescribed focal test NV |
| *Unsupervised learning-based registration* | | | | | |
| VoxelMorph | 2019; IEEE TMI [48] | Image pair; unsupervised (optional auxiliary segmentations) | Amortised similarity-plus-smoothness objective | General registration datasets; accuracy and run time versus selected comparators | NV |
| TransMorph | 2022; Med Image Anal [49] | Volumetric pair; unsupervised | Hybrid Transformer–CNN; diffeomorphic and Bayesian variants | Brain MRI (inter-patient, atlas) and phantom-to-CT; topology preservation; uncertainty calibration | NV for cardiac pathology |
| Fourier-Net | 2023; AAAI [50] | Image pair; unsupervised | Band-limited base representation; inverse Fourier decoder; diffeomorphic integration changes final spectral support | Registration benchmarks; accuracy with speed, memory and multiply-adds | NV |
| Fourier-Net+ | 2026; IEEE TNNLS [51] | Band-limited image representation; unsupervised | As Fourier-Net with shortened contracting path; cascaded variant | Five datasets including one 3D cardiac MRI set; Dice, Hausdorff distance, ED volume and EF versus TransMorph and other baselines at lower memory and compute (abstract level) | NV; cardiac evaluation is geometric/functional, not regional strain |
| *Cardiac-specific estimators* | | | | | |
| Joint motion and segmentation | 2018; MICCAI [52] | Cine sequence; unsupervised motion, weakly supervised segmentation | Shared multi-scale encoder; Siamese recurrent spatial transformer | 220 total subjects, including 100 test subjects; mask propagation and speed versus competing methods | NV |

| Method | Year; source | Input; supervision | Prior or representation | Reported data and validation | Evaluation on pathology or focal abnormality |
|---|---|---|---|---|---|
| Apparent flow | 2019; Med Image Anal [53] | Two cine phases; semi-supervised | Pixel-wise flow combined with masks | ACDC; nine shape/volume/EF and motion features; 95% five-fold cross-validation and 94% held-out challenge test accuracy | Pathology classes are the target; motion validated via the downstream label |
| Motion Pyramid Networks | 2020; MICCAI [54] | Cine pair; unsupervised, cyclic teacher–student | Multi-scale fused pyramid | Two public datasets; mask metrics and errors against known motion applied by a separate network; inference time | Model-derived imposed motion; not independent physiological truth |
| FOAL | 2020; CVPR [55] | Cine sequence at test time; meta-learned online optimiser | Test-time adaptation of a learned tracker | Two public datasets; accuracy under train–test mismatch; about 0.4 s per video for online optimisation | Distribution shift is the target, not characterised as focal abnormality |
| MulViMotion | 2022; IEEE TMI [56] | SAX plus 2-/4-chamber LAX cine; unsupervised with weak shape supervision | Hybrid 2D/3D network; shape regularisation module | 580 UK Biobank subjects: 450 training, 50 validation, 80 test; 3D LV motion comparison | NV |
| DeepMesh | 2024; IEEE TMI [57] | SAX and LAX cine; unsupervised via differentiable rasteriser | Template 3D endo/epicardial mesh propagated to subject space | UK Biobank; vertex-wise 3D displacement with temporal correspondence and common template indexing | NV |
| Bidirectional registration network | 2023; Comput Biol Med [58] | Cine sequence; unsupervised | Bidirectional reference-to-frame prediction with sequence context | Public cine data (abstract level) | NV |
| Low-rank groupwise deformations | 2024; preprint (MSc thesis) [59] | Image sequence; unsupervised | Groupwise diffeomorphic registration towards a low-rank aligned-image stack | NV; comparison of resulting low-rankness with other groupwise approaches | NV |
| MCM (Mamba) | 2025; MICCAI workshop [60] | Cardiac-cycle image sequence; unsupervised | Bidirectional Mamba block; adjacent-frame motion decoder | Two public datasets; comparison with conventional and learned trackers | NV |
| SegMorph | 2025; IEEE TMI [61] | Cine sequence; joint motion and segmentation | Recurrent variational representation | Public cine data (abstract level) | NV |
| CardioMorph-Net | 2026; Med Image Anal [62] | SAX cine volumes; recursive registration of segmentation maps without an intensity loss | Recurrent VAE with separate segmentation and motion posteriors; uncertainty map | UK Biobank and M&Ms; warped versus reference masks; uncertainty versus other probabilistic methods | M&Ms is multi-centre/multi-pathology, but the comparison is mask agreement, not focal motion accuracy |
| Bidirectional latent neural ODE | 2026; Comput Med Imaging Graph [63] | Cine sequence; unsupervised | Continuous-time latent ODE; Lagrangian consistency | ACDC and M&Ms; temporal consistency and plausibility versus selected comparators | NV |
| STCMT-Net | 2026; Med Image Anal [64] | 4D CT and cine MRI; unsupervised | Unsupervised keypoints as anchors; dense field from matched-keypoint basis vectors | 4D CT and cine datasets; comparison with competing methods; strain analysis separating cardiac conditions | Group separation by condition; prescribed focal test NV, material-motion reference NV |
| Semi-supervised motion flow | 2025; Comput Biol Med [65] | Full cine sequence; base, distance-map and fully supervised configurations | Memory network over all frames | Abstract: 271 patients; EPE 1.02 mm base, 0.95 mm distance-map, 0.91 mm full supervision; RAFT 1.19 mm; global peak RS/CS correlation 0.83/0.90 | EPE reference: manually initialised contours propagated by conventional FT; not dense material truth; focal endpoints NV |

| Method | Year; source | Input; supervision | Prior or representation | Reported data and validation | Evaluation on pathology or focal abnormality |
|---|---|---|---|---|---|
| IG-BiReg | 2026; Phys Med Biol (online first) [66] | Input/supervision details NV | Title-level implicit-guided bidirectional residual registration | Primary scientific content not accessed in finalisation; performance NV | NV |
| CMR foundation models | 2025–2026 [67, 68] | 36M images/27,524 subjects (self-distillation); 15M cine images/74,916 studies | Pre-trained representation fine-tuned per task | Segmentation, landmark localisation, diagnosis, prognostication across independent datasets | Dense motion was not among the tasks in the retrieved summaries; other task validation was reported |
| *Physics-, biomechanics- and representation-constrained estimators* | | | | | |
| Biomechanics-informed prior | 2023; Med Image Anal [69] | Cine sequence; unsupervised, prior learned from simulated deformations | VAE manifold of plausible deformations traversed at tracking time | Three public cine datasets; tracking accuracy, volume preservation, generalisability, derived strain | NV as a prescribed focal experiment |
| WarpPINN | 2023; Med Image Anal [70] | Cine bSSFP; per-case optimisation | Neural deformation with hyperelastic regulariser and Jacobian near-incompressibility; Fourier features | Synthetic examples; 15-volunteer cine benchmark; landmark tracking; strain by automatic differentiation | Healthy benchmark; focal abnormality not evaluated |
| WarpPINN-fibers | 2026; Eng Comput [71] | Cine MR; per-case optimisation | As WarpPINN plus fibre-stretch penalty along synthetic fibres | Known-motion synthetic phantom; 15 healthy participants: landmark comparison and strain-curve plausibility without reference strain curves | Healthy benchmark; focal abnormality not evaluated |
| Implicit neural representation | 2026; STACOM [72] | NV | Implicit neural representation of intramyocardial motion and strain | NV (no abstract in the record retrieved) | NV |
| Space–time finite-element correlation | 2026; Med Image Anal [73] | Routine SAX and LAX cine; correlation framework | Region-specific biomechanical regularisation; data-driven temporal decomposition; multi-view alignment | 2D+$t$: synthetic known motion/strain, three public datasets and one clinical dataset; 3D+$t$: clinical masks only; two FT and four DL comparators (abstract) | One synthetic ground-truth setting; in-vivo references are landmarks and masks; size/severity sweep NV |
| *Reference-supervised and end-to-end strain* | | | | | |
| Cine-only infarct delineation | 2018; Med Image Anal [74] | Non-contrast cine; supervised on manual LGE delineation | LSTM motion features plus optical flow, fused per pixel | 165 cine datasets; pixel accuracy 95.0%, Dice 89.9%, $\kappa = 0.91$ against two radiologists | Infarction is the target; reference is LGE tissue, not motion |
| Survival-supervised motion | 2019; Nat Mach Intell [75] | RV cine-derived motion; supervised on survival in chronic thromboembolic pulmonary hypertension | Learned low-dimensional motion representation | 302 patients; internally bootstrap-corrected Harrell C 0.75 (95% CI 0.70–0.79) versus conventional manual-variable model 0.59 (0.53–0.65) | Prognostic discrimination in chronic thromboembolic PH; not an LV focal-motion or management-impact endpoint |
| DeepTag | 2021; CVPR [76] | Tagged MR (comparator acquisition); unsupervised | Bidirectional diffeomorphic registration; differentiable composition to Lagrangian field | Clinical tagging dataset; landmark tracking accuracy and efficiency | NV |
| 3D tagging networks (synthetic training) | 2025; JCMR [77] | 3D tagged MR; synthetic training from a biophysical LV model | Two networks predicting LV motion through the cycle | Synthetic validation plus external human manual-landmark comparison and porcine strain comparison with manual cine annotations | Tagging analysis, not routine cine; manual reference has its own uncertainty |

| Method | Year; source | Input; supervision | Prior or representation | Reported data and validation | Evaluation on pathology or focal abnormality |
|---|---|---|---|---|---|
| DeepStrain | 2021; Front Cardiovasc Med [1]; 2023 comparison (online 2022) [2] | Cine bSSFP; image, anatomical-label and smoothness losses | Segmentation plus motion network; end-diastolic reference; Green–Lagrange strain | ACDC ($n$ = 150); CMAC landmark EPE 2.9 ± 1.5 mm; scan–rescan in ten volunteers; 47 healthy/533 patients versus manual FT, $r$ 0.83–0.85 radial and 0.88–0.91 circumferential | Independent MRXCAT2.0 evaluation on normal, dilated, hypertrophic and infarcted cases (main text, known-motion tests) |
| StrainNet | 2023; Radiol Cardiothorac Imaging [4] | Contour time series (DENSE-magnitude at displacement test; cine FT at strain test); supervised on DENSE displacement | 2D+$t$ CNN; prior implicit in DENSE-derived training distribution | 243 development/62 internal-test participants (670/190 sections); EPE 0.75 ± 0.35 mm with DENSE-magnitude contours; cine Ecc ICC 0.87 global/0.75 segmental versus FT 0.72/0.48; segmental LoA −9.16 to 8.07 percentage points in 59 | Mixed patient population with material-sensitive reference; size/severity sweep NV; no focal location/extent endpoint |
| DL synthetic strain | 2023; Radiol Cardiothorac Imaging [78] | Cine at inference; training targets: co-registered 4D-flow velocity and commercial-software segmentation | Velocity and segmentation network; strain derived from estimated motion | Development: 223 exams/219 patients; clinical test: 53 patients/846 segments, four-reader WMA consensus; ROC AUC 0.90; 30% peak Err threshold: sensitivity 86%, specificity 85% | WMA clinical endpoint; reader labels evaluate the output, not the motion-training target |
| DENSE-guided rotation network | 2025; IEEE ISBI [79] | Routine bSSFP cine; supervised on DENSE displacement | Separate rotation and radial branches fused | NV in the record retrieved; strain accuracy against DENSE with large rotation as target | NV |
| DL aligned strain | 2025; Radiol Artif Intell [80] | Cine key phases and segmentation; deformation loss: SSIM, smoothness, Dice | Deformation-derived strain aligned for downstream classification; LGE labels classify patients | 139 DMD patients: 57 development, 82 same-centre temporal holdout; endpoint: at least one LGE-positive segment | Patient-level fibrosis classification, not scar-boundary error or LGE-supervised motion |
| MyoNet | 2026; Bioengineering [81] | Binary spatiotemporal inputs from cine segmentation; displacement targets from SinMod on paired tagging | CNN alternating spatial and temporal convolution | Dahl salt-sensitive rats after radiation; ICC 0.973 circumferential, 0.975 radial | Preclinical tagging-reference agreement; not human cine validation |

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